

Foldable Long-Distance Ball Transfer Mechanism - Final Report

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

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Project	Foldable Long-Distance Ball Transfer Mechanism
Course	Machine Design Theory
Document Type	Final machine design report
Primary Role	Team lead, concept designer, and fabrication planner
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Executive Summary

This project designed a deployable ball-transfer mechanism under strict packaging and reliability constraints. The mechanism had to move from compact storage to working configuration and transfer a ball across distance.

The final design used fabrication-aware choices such as 3D-printed parts, MDF structure, hinges, and latch hardware to improve deployment repeatability, locked-state stiffness, and repair speed.

The final mechanism performed reliably during evaluation and strengthened my ability to connect mechanism architecture with fabrication, assembly order, and test behavior.

Applicant Contribution Signals

- Treated packaging as the central engineering problem rather than a late detail.
- Balanced ideal motion paths with buildable mechanics and available fabrication methods.
- Used accessible hardware to improve reliability and iteration speed.
- Led concept selection, folding architecture, fabrication planning, and final refinement.

Mechanism Architecture

The mechanism needed to satisfy storage constraints, deployment motion, stiffness, ball-transfer path, and repair access simultaneously.

This shifted the project from simply making a transfer path to designing a compact machine that could repeatedly move between storage and operation.

Fabrication Strategy

Early folding concepts were simplified and adapted to the tools and hardware available.

Hinges, latch hardware, MDF, and 3D-printed parts were chosen to create a more predictable motion path and locked state.

Validation

Final testing emphasized repeatable deployment and practical reliability rather than unnecessary complexity.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Strict storage envelope	Made folding architecture central	Prevented a working path from becoming impossible to package	Design for constraints
Repeatable deployment	Used hinge and latch hardware	Improved motion predictability and locked stiffness	Mechanism design
Fabrication limits	Used accessible materials and repair-friendly assembly	Made iteration faster	Fabrication-aware design
Final reliability	Prioritized stable execution over complexity	Improved final-test performance	Engineering judgment

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.